

Indian Institute of Engineering Science and Technology, Shibpur
B.Tech. 3rd Semester Final Examination, November, 2018
Discrete Structures (CS 303)

Full marks: 70

Time: 3 hours

Attempt any five from question number 1 to question number 7, and any two from question number 8 to question number 10.

1. a) Define the following terms:

(i) Partial Order Relation

(ii) Chain and Antichain

(iii) Complemented Lattice

(iv) Boolean Lattice

b) Let $(A, \leq)$ be a Partially Order set. Suppose the length of the longest chains in A is m . Then prove that the elements in A can be partitioned into m disjoint antichains.

(4+6)

2. a) In a distributive lattice, if $x \wedge \bar{y} = 0$, then prove that $x \leq y$. The symbols have their usual meanings.

b) For any a, b, c, d in a lattice $(A, \leq)$, if $b \leq a$ and $d \leq c$ then prove that

(i) $b \vee d \leq a \vee c$ and

(ii) $b \wedge d \leq a \wedge c$

(5+5)

3. a) Solve the difference equation, $u_r = \sqrt{u_{r-1} + \sqrt{u_{r-2} + \sqrt{u_{r-3} + \dots}}}$, given that $u_0 = 4$.

b) Let e_n be the number of edges in a complete graph of n vertices.

(i) Derive a recurrence relation for e_n in terms of e_{n-1} .

(ii) Solve the recurrence relation.

(5+5)

4. a) Find out the particular solution of the difference equation $a_r + a_{r-1} = 3r2^r$

b) Define the convolution of two numeric functions a and b . Compute it when $a = 3^r; r \geq 0$ and $b = 2^r; r \geq 0$.

c) Use generating function to evaluate the sum $1^2 + 2^2 + 3^2 + \dots + n^2$ (3+3+4)

5. a) Using numeric function, determine the number of ways in which r objects can be selected from n objects with unlimited repetitions.

b) Derive only the recurrence relations for solving the following problems:

(i) Suppose a coin is tossed r times. You want to determine the number of sequences of outcomes in which heads never appear on successive tosses.

(ii) Consider a certain nuclear reaction inside of a reactor containing nuclei and high and low energy free particles. There are two kinds of events:

- A high-energy particle strikes a nucleus and is absorbed, causing it to emit two high energy particles and three low-energy particles; and
- A low-energy particle strikes a nucleus and is absorbed, causing it to emit two high energy particles and one low-energy particle.

You want to determine the number of high-energy and low-energy particles in the system at r -th μs .

Assumptions: (I) Every free particle causes an event $1 \mu s$ after it is emitted. (II) A single high energy particle is injected at time 0 into a system containing only nuclei. (5+(2+3))

6. a) Define: term, atom and well formed formula (wff) in First Order Predicate Logic (FOPL) and give suitable example of each of the above.

b) Translate the following basic axioms of whole numbers into formulas:

- (i) For every number, there is one and only one immediate successor
- (ii) There is no number for which 0 is the immediate successor
- (iii) For every number, other than 0, there is one and only one immediate predecessor

c) Obtain Prenex Normal Form (PNF) of the following formula:

$$\forall x \forall y (\exists z P(x, y, z)) \wedge \exists x \exists u (Q(x, u) \rightarrow \exists y \exists w Q(y, u)) \quad (3+3+4)$$

7. a) Categorize the following well formed formulas (wffs) into valid, invalid, satisfiable, or unsatisfiable.

- (i) $\neg ((p \rightarrow q) \wedge (q \rightarrow r) \rightarrow (q \rightarrow r))$
- (ii) $(p \rightarrow (\neg q \rightarrow r)) \wedge (p \rightarrow \neg q) \rightarrow (p \rightarrow r)$

b) Solve the following problem using the concept of logical consequence of a formula to the other formulas, without using truth Table.

Problem: Given that if the congress refuses to enact new laws, then the strike will not be over unless it lasts more than one year and the President of the firm resigns. Will the strike not be over if the congress refuses to act and the strike just starts?

c) Justify whether the following arguments are valid.

- (i) All men are mortal. Confucius is a man. Prove that Confucius is mortal
- (ii) If Sita was at the fair, then her father was negligent or her mother was not at home. If her mother was not at home, then her father was not negligent. Her mother was at home. Therefore, Sita was at the fair. (3+4+3)

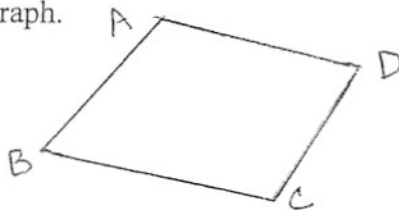
8. Prove that in a binary tree

- (i) number of vertices is always odd; and
- (ii) number of pendant vertices is one more than the number of internal vertices.

(2+2)

b) Define chromatic number and chromatic polynomial of a graph. State the decomposition theorem and apply it to find the chromatic polynomial and chromatic number of the following graph.

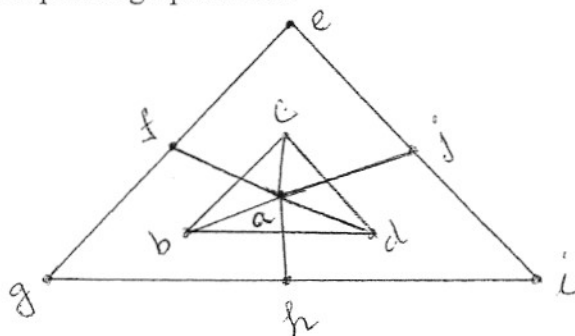
(2+(1+2+1))



9. a) Prove the Euler's Formula that a connected graph with n vertices and e edges has $e - n + 2$ regions.

b) Write an algorithm to detect the planarity of a graph and apply it to determine if the following graph is planar graph or not.

(4+6)



10. a) Define fundamental circuit and spanning tree of a connected graph. How many fundamental circuits does a graph have? Explain.

(2+2)

b) Write the PRIM's Algorithm and apply it on the following graph to determine a minimal spanning tree.

(6)

